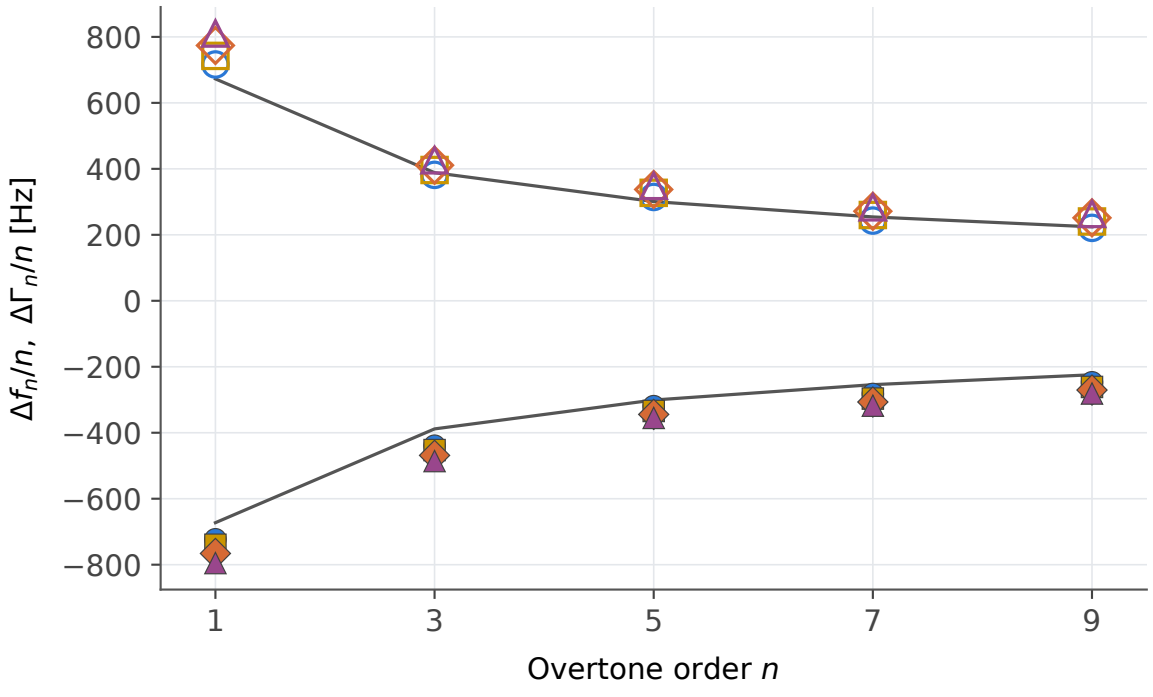
Glucose series · DS-3 · phase-shifted Lorentzian

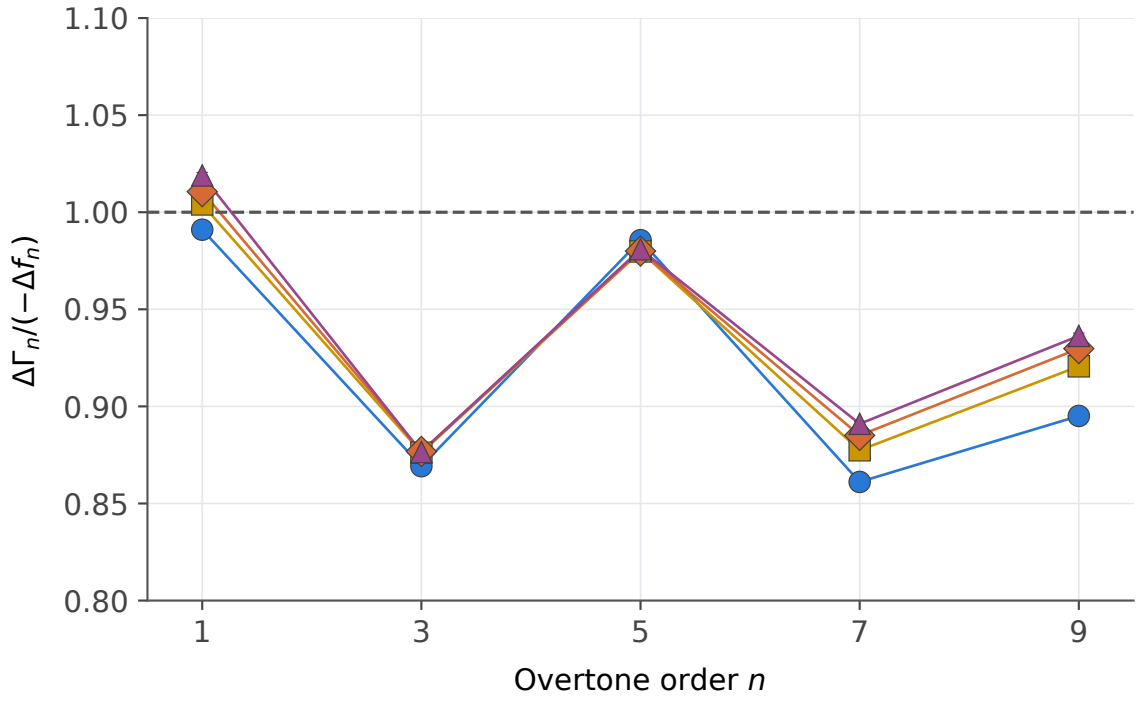
Water Glucose 5% Glucose 7.5% Glucose 10%

Concentrations in % w/v. T5: filled = frequency; open = half-bandwidth; line = KG for water at 25 °C.

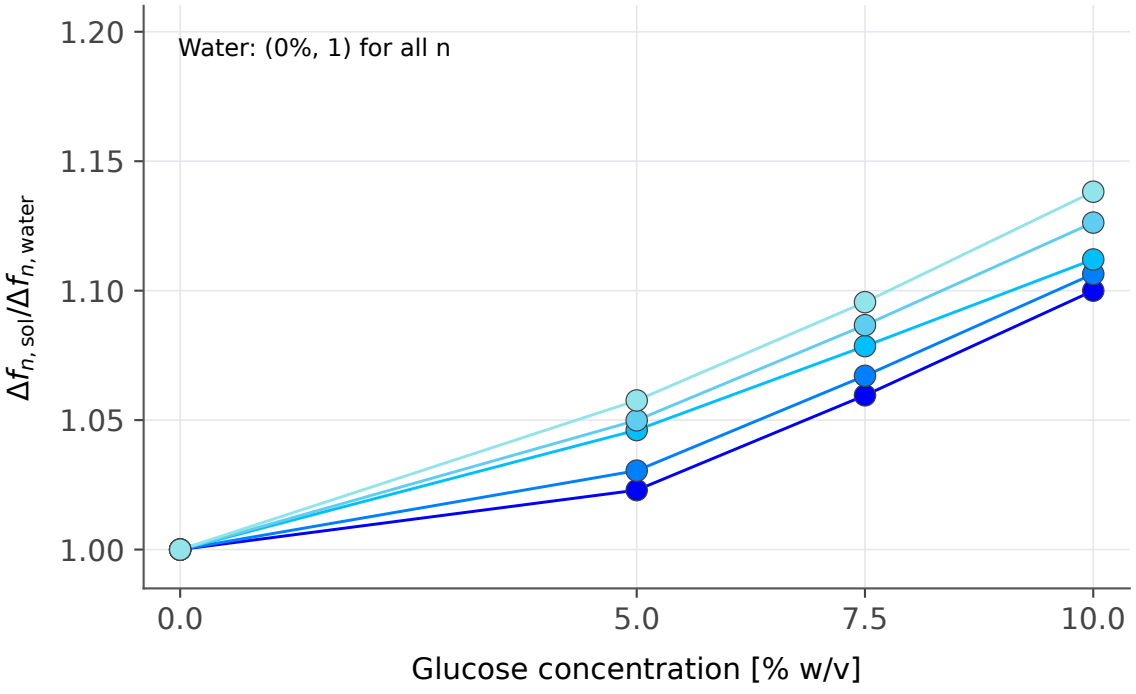
(a) T5 · signed normalised shifts



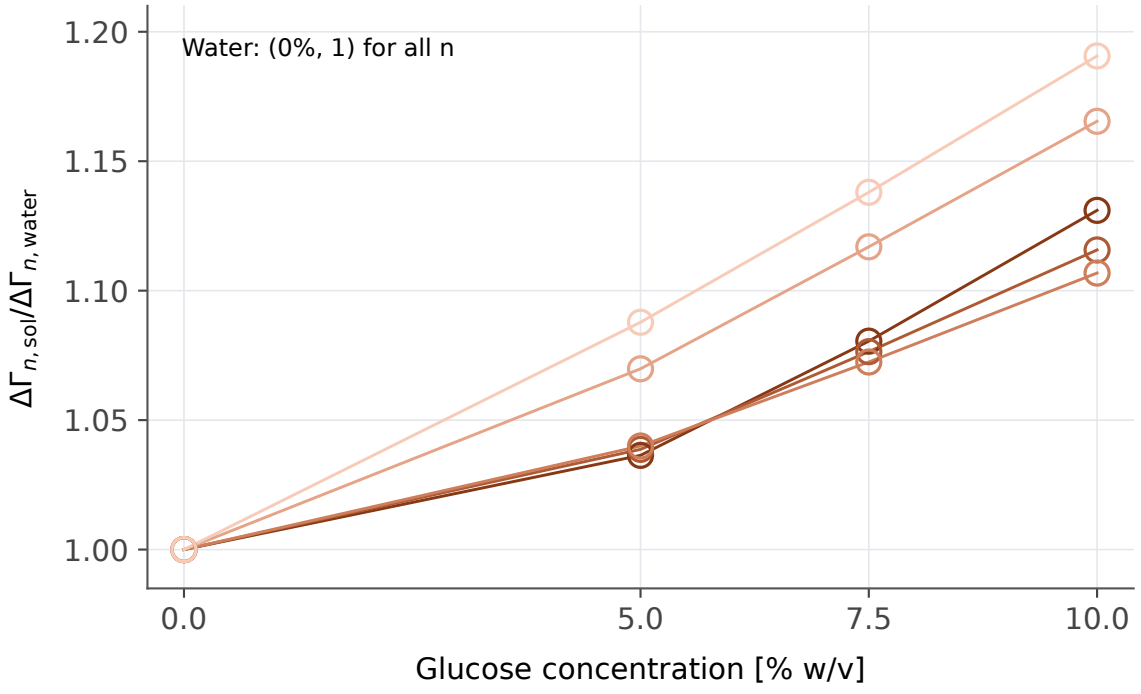
(b) KG-1 · Newtonian ratio



(c) Frequency · relative to water



(d) Half-bandwidth · relative to water



$R_{f,n} = \Delta f_{n,\mathrm{sol}}/\Delta f_{n,\mathrm{water}}$ $n=1$ $n=3$ $n=5$ $n=7$ $n=9$

$R_{\Gamma,n} = \Delta\Gamma_{n,\mathrm{sol}}/\Delta\Gamma_{n,\mathrm{water}}$ $n=1$ $n=3$ $n=5$ $n=7$ $n=9$

Panels (c, d): air-referenced shifts divided by the water shift at the same overtone. Lines connect measurements as guides to the eye.